

are haggled by the validators. A majority of the validators need to sign the square, before it is added to the chain. Along these lines, there will be 'no forks or returns'. When a square is submitted, the state is conclusive. The convention takes into consideration 'sub second square occasions'. New squares are however just made within sight of pending exchanges.

- **Native contracts:** HydraChain gives a framework to create shrewd agreements in the Python elevated level language. The advantages are fundamentally decreased improvement times and better investigating capacities. As the Ethereum virtual machine is circumvented, local agreement execution is also much quicker. Local contracts bolster the ABI and are operable with EVM (Ethereum virtual machine) based agreements written in the Solidity or Serpent dialects, and can exist together on a similar chain. The imperative that all validators need to run a hub designed with a similar arrangement of local agreements is quite sensible in private chain settings.
- **Customisability:** Numerous parts of the framework can be unreservedly arranged to fit custom needs. For instance, exchange expenses, gas limits, beginning distribution, square time and so on can be balanced without too much of a stretch.

- **Easy deployment:** Setting up a test system should be possible with almost zero designs.

The specific features of HydraChain are:

- (a) Ledger support
- (b) Smart contract support
- (c) Private blockchain support
- (d) Network security
- (e) Processing speed

HydraChain's upcoming features are:

- (a) Documentation
- (b) Proof of identity - KYC/AML (anti-money laundering)
- (c) Selective state sharing
- (d) Chain interoperability

The 'Hydra' in the name indicates that the product will support the running of a hub that simultaneously takes an interest in different chains. Next to other applications, this helps to support cross chain asset transfers as a native feature.

The architecture of HydraChain

HydraChain is a private blockchain that is dependent on Ethereum, an open blockchain. It uses PBFT to keep up the consistency of records at hubs in the square chain as opposed to a mining system, and its speed is a lot quicker than blockchains that use the mining procedure. Figure 1 shows the HydraChain design.

The installation process

While heading off to the execution side,

HydraChain can be downloaded and introduced from GitHub, and offers a simple arrangement of systems. For system organisation, a few Docker document layouts are currently accessible in HydraChain. These layouts can be used while we make new systems.

You can install dependencies on Ubuntu as follows:

```
> sudo apt-get install libssl-dev
libffi-dev libtool python-dev autoconf
automake
```

For installation, use the following code:

```
> git clone https://github.com/
HydraChain/hydrachain
> cd hydrachain"
> python "setup.py" develop
```

For multiple nodes in a single process, use the command given below:

```
> hydrachain -d datadir runmultiple
--num_validators=3 --seed=42
```

For multiple nodes in multiple processes on the same machine, use the following command:

```
> hydrachain -d <datadir> rundummy
--num_validators=3 --seed=42 --node_
num=0
```

To run three nodes in three terminals, use the following command:

```
> hydrachain -d datadir0 rundummy
--node_num=0
> hydrachain -d datadir1 rundummy
--node_num=1
> hydrachain -d datadir2 rundummy
--node_num=2
```

The *rundummy* command automatically configures a setup for *num_validator* nodes (instances of the application) which are running on the same machine. The node ID of each

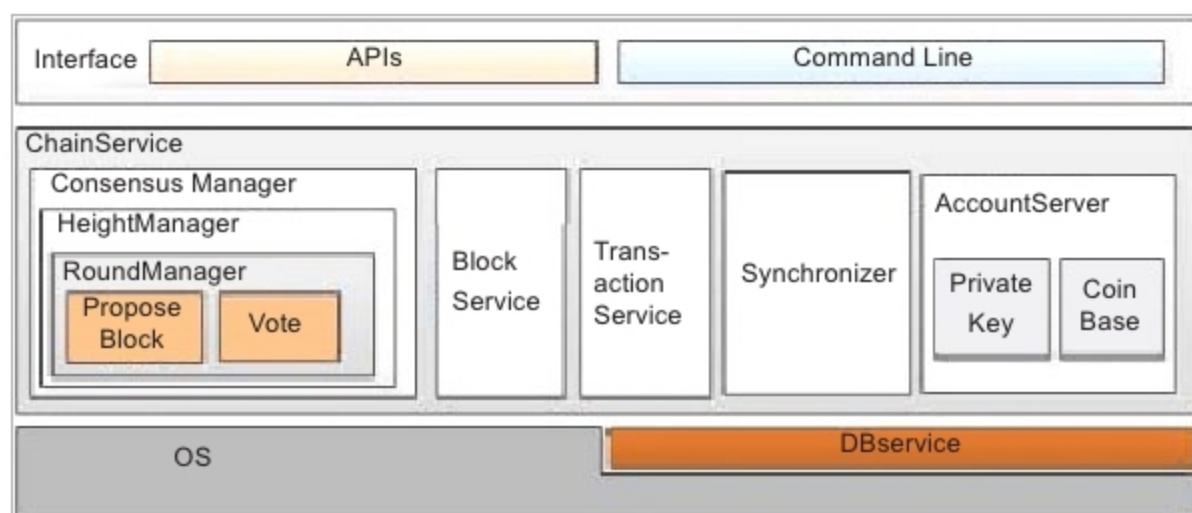


Figure 1: HydraChain architecture